

1. THE SAMPLING FRAME

The sampling frame is the list of units from which a sample is actually drawn. Any mismatch between the frame and the target population produces coverage error, and coverage error cannot be repaired by weighting after the fact. Frames for household surveys in India are commonly built from Census enumeration blocks in urban areas and from village lists in rural areas.

2. STRATIFICATION

Stratified random sampling divides the population into homogeneous strata and draws an independent sample within each stratum. When the strata are internally homogeneous with respect to the study variable, stratification reduces the variance of the estimator relative to simple random sampling of the same size. Allocation across strata may be proportional to stratum size, or optimal, in which case it also accounts for within-stratum variance and cost of enumeration.

3. PROBABILITY PROPORTIONAL TO SIZE

Probability proportional to size selection gives larger units a greater chance of inclusion. This improves efficiency when the study variable is correlated with the measure of size used for selection. In multi-stage designs, first-stage units such as villages or urban blocks are frequently selected with probability proportional to population, with a fixed number of households selected at the second stage.

4. WEIGHTING AND ESTIMATION

The design weight of a unit is the reciprocal of its probability of selection. Design weights are adjusted for non-response and are often calibrated to known population totals so that survey estimates agree with external benchmarks such as projected population. Variance estimation must reflect the design: treating a multi-stage stratified sample as if it were simple random sampling understates standard errors, sometimes severely.

5. NON-SAMPLING ERROR

Non-sampling error includes measurement error, processing error and non-response. In large official surveys non-sampling error frequently exceeds sampling error in magnitude, and unlike sampling error it does not shrink as the sample grows. It is controlled through interviewer training, instrument design, supervision and independent re-interview of a subsample.

6. DATA QUALITY, EDITING AND IMPUTATION

Editing rules detect records that violate logical or arithmetic constraints. Selective editing concentrates limited review effort on the records whose correction would most change the published aggregate. Imputation replaces a missing value with a plausible substitute derived from responding units. The imputation method must be recorded, because imputed values add uncertainty that naive variance estimates ignore. Donor-based hot-deck imputation preserves the distribution of the variable better than mean imputation, which artificially reduces variance.

7. QUESTIONNAIRE AND FIELD OPERATIONS

A well designed questionnaire places sensitive questions late in the interview, after the respondent has committed time to the exercise and rapport has been established. Computer assisted personal interviewing enforces skip patterns and range checks at the point of collection, which removes an entire class of downstream editing work and shortens the interval between collection and release.